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Palaeomagnetic results from the Borrowdale Volcanic Group, English Lake District

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Summary. Palaeomagnetic results are reported from 47 sites within the Borrowdale Volcanic Group of late Llanvirnian–Llandeilian age. The mean stable remanence direction is $D = 329°$, $I = -49°$ ($\alpha_{95} = 7°$) after alternating field demagnetization and correction for tectonic dip. The corresponding palaeomagnetic pole position is 0°, 203° E. This pole differs from both older and younger Lake District Ordovician poles but the differences are more likely to arise from local geological complexity, sampling problems and the limitations of available statistical methods than from true polar wander. It is suggested that no systematic polar movement relative to the British Isles can be inferred during Ordovician time. An overall Ordovician pole position for the British Isles is calculated from the present result and eight others to be in the vicinity of 7° N, 190° E ($\alpha_{95} = 9°$).

Introduction

The Borrowdale Volcanic Group comprises a thick sequence of calcalkaline lavas and pyroclastic rocks of late Llanvirnian–Llandeilian age, which outcrop extensively on the southern limb of the Lake District anticline. These rocks are younger than, and chemically distinct from, the Eycott Group, formerly known as the 'Northern Borrowdale Volcanics' of lower Llanvirnian age, which appear on the northern limb (Downie & Soper 1972).

Briden & Morris (1973) reported palaeomagnetic results from the Eycott Group and from the Carrock Fell gabbro–granophyre complex which is intruded along the junction between the Eycott Group and the Skiddaw Slates. For the Eycott Group the mean stable remanence direction was $D = 0°$, $I = -45°$ ($\alpha_{95} = 5°$) which corresponds to a palaeomagnetic pole position of 7° N, 177° E. The Carrock Fell Complex gave $D = 350°$, $I = -22°$ ($\alpha_{95} = 14°$), and a pole position of 28° N, 188° E. Both these sets of results agree closely with those from the Ordovician Builth Volcanics of Wales and the Aberdeenshire Gabbros (Briden, Morris &

Piper 1973). Recent results from the Ordovician of North Wales (Thomas & Briden 1976) suggest that the mid and late Ordovician field was more complex than was previously realized. Further evidence is sought through the present study, which concentrates on the Borrowdale Volcanic Group in the south-western and eastern parts of the Lake District, not previously sampled for palaeomagnetic purposes.

Geology and sampling

The Borrowdale Volcanic Group includes lava flows, ash flows, airfall and detrital tuffs, ranging from basic andesitic to rhyolitic in composition. It rests unconformably on Llanvirnian Skiddaw Slates eastward from Keswick (Wadge 1971). In the south-west, where the succession is about 7 km thick (Numan 1974), marine pelites are interbedded with the lowest 2 km of the volcanic succession and there may be a broadly conformable transition between the Skiddaw Slates and volcanic rocks within the Llanvirnian (N. J. Soper, private communication). The Borrowdale Volcanics are overlain unconformably by marine sediments of Caradocian age, the Coniston Limestone Group, of which the oldest (the Stile End Beds) are Actonian. As a younger limit for the age of the Borrowdale Volcanics this is an underestimate, since the volcanic rocks underwent folding and erosion in pre-Actonian time. It is therefore considered that the bulk of the Borrowdale Volcanic Group was erupted in Llandeilian time, with the lower part of the succession in the west probably older (perhaps late Llanvirnian) than the lowest part in the east. The Borrowdale Volcanics were folded into a series of open folds with ENE axial trend (Moseley 1972). Soper & Numan (1974) confirmed that two of the regional-scale folds (the Ulpha Syncline and the Nan Bield anticline) are of pre-Caradocian age. The rocks were subjected to much greater strain in late Silurian time during the main phase of the Caledonian orogeny. This produced penetrative slaty cleavage in the tuffs but brittle failure in the more competent lavas and welded tuffs.

Massive, uncleaved andesites were sampled by drilling cores directly in the field. Bearings were taken with a sun compass whenever possible, otherwise with a magentic compass. Local dip was estimated from the nearest bedded tuffs to the sites, in order to apply tilt corrections to the remanence directions. The localities visited are shown in Fig. 1. In the east (~ latitude 54.5° N, longitude 2.8° W), samples were obtained from the Haweswater area (mapped by Nutt 1968), from the north-west of Ullswater (Moseley 1964), from Kentmere and from the head of Longsleddale (Soper & Numan 1974). The Haweswater sites lie on the opposite (northern) limb of the Nan Bield anticline from those in Kentmere and Longsleddale; the dip at the Ullswater sites is determined by the Ullswater anticline. Three sites were sampled in the Haweswater dolerite, an intrusion interpreted by Nutt (1968) as a vent related to the Borrowdale Volcanics. In the west (~ latitude 54.3° N, longitude 3.2° W), samples of the Ulpha Andesite and the Lower Andesite of Firman (1957) were taken from Dunnerdale and the Ulpha Fells respectively. Here, the volcanics are present in both limbs of the Ulpha syncline; on its southern limb they show a change in strike due to the younger Black Combe anticline (Helm 1970).

The intensities and remanence directions of some of the samples were measured on a PAR spinner magnetometer, the remainder on a Digico balanced fluxgate spinner magnetometer. Alternating field (af) demagnetization techniques were applied to test the stability of the remanence. One pilot sample from each site was subjected to progressive demagnetization at increasing peak fields. The stable remanence was taken as the state where the magnetic vector showed least change during successive demagnetizing steps according to the stability index of Briden (1972). Having selected an optimum peak field for each site, all samples were 'cleaned' by af demagnetization.

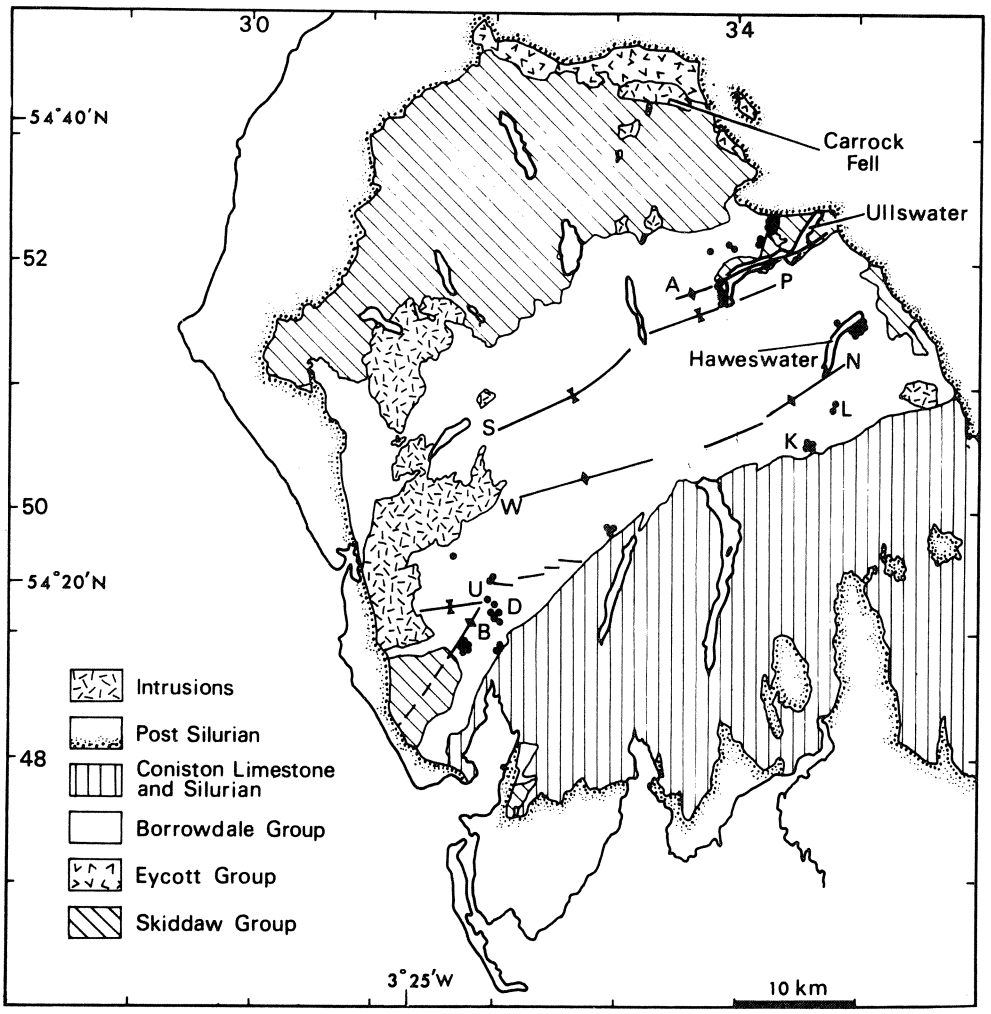


Figure 1. Geological map of the Lake District showing axial traces of major folds and palaeomagnetic sampling localities (dots) listed in Tables 1 and 2. D – Dunnerdale, K – Kentmere, L – Longsleddale, A – Ullswater anticline, S–P – Scafell–Place Fell syncline, W–N – Wrynose–Nan Bield anticline, U – Ulpha syncline, B – Black Combe anticline.

Results

The results are presented in Tables 1–3 and Figs 2–4. All the western sites had the same NRM polarity. Fourteen out of 19 western sites had total NRMs which were significant (in the sense of being non-random with 95 per cent probability (Watson 1956b)) and this was increased to 17 by af cleaning (Table 1). At 12 of the sites the within-site precision estimate, k, was appreciably greater after cleaning. Some pilot specimens showed noticeable change of direction on cleaning at low peak fields – Fig. 3(c) – others showed only a gradual decrease of intensity on progressive demagnetization – Fig. 2(a). Median destructive fields ranged between 25 and 60 mT. Usually the directions were very stable at peak fields above 30 mT. It was not possible to carry out a specific fold test on the western sites by sampling across the Ulpha syncline, as no suitable outcrop could be found on its northern limb. However,

Table 1. af cleaned results from the western Borrowdale Volcanics.

Site	National Grid reference (100 km square SD)	Peak field (mT)	N	R	k	α_{95}	D (*in situ*)	I (*in situ*)	D (dip corrected)	I (dip corrected)
7 W	201917	20	5	4.93	51.7	10	42	−58	323	−60
10 W	199918	30	5	4.98	200.3	5	20	−61	317	−49
11 W	201911	30	5	4.95	74.2	9	60	−54	332	−71
12 W	199925	30	4	3.99	568.5	4	25	−49	334	−48
13 W	198927	40	4	not significant						
15 W	202913	30	4	3.66	8.7	33	69	−69	288	−66
9 W	175890	30	4	3.67	9.7	31	352	−30	307	−48
17 W	174892	30	4	3.70	10.0	30	12	−32	315	−64
18 W	175892	40	5	not significant						
19 W	175891	30	4	3.97	85.8	10	20	−47	273	−71
20 W	172893	30	5	4.97	118.6	7	16	−55	258	−65
22 W	239983	20	6	5.90	52.0	9	2	−66	324	−29
23 W	239983	20	6	5.98	213.2	5	17	−55	337	−26
24 W	196944	20	7	6.99	287.3	4	113	−73	4	−40
25 W	295980	20	6	5.93	71.8	8	111	−35	11	−38
26 W	293983	20	6	5.79	23.6	14	135	−32	308	−66
27 W	295984	20	5	4.45	7.2	30	113	−33	348	−59
28 W	166962	20	6	4.77	17.5	19	231	−59	277	−35
29 W	208890	20	4	4.00	327.4	14	346	−81	321	−37

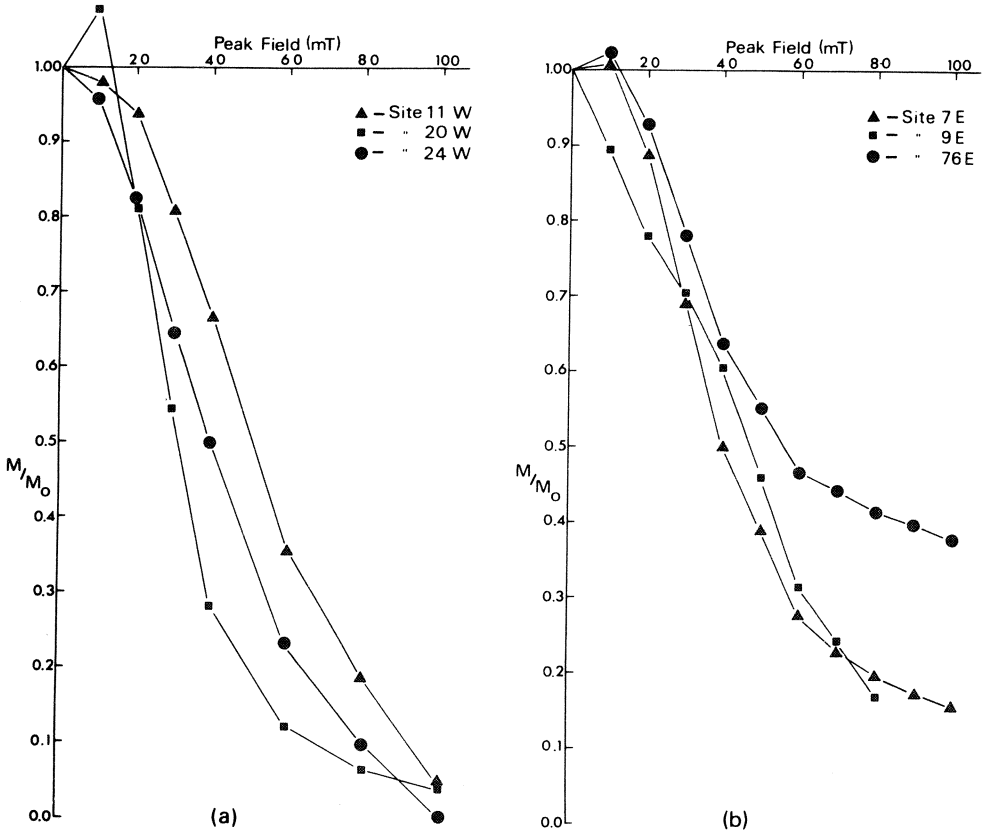


Figure 2. Examples of intensity change of progressive af demagnetization shown by pilot samples from the sites indicated (a) western sites, (b) eastern sites.

the precision of site means was significantly greater with 99 per cent probability (Cox 1969) after the stable *in situ* directions had been corrected for tilt, using local dip measurements and simple rotation about the strike. Combining the cleaned and dip-corrected site mean directions gave $D = 322°$, $I = -55°$ ($\alpha_{95} = 11°$) – Fig. 3(b).

Twenty-two out of 28 eastern sites had total NRMs significant at the 95 per cent level: this number was not increased by af cleaning. The eastern results fall into a principal group and a smaller number of anomalous sites (Table 2 and Fig. 3). The principal group of 14 sites is centred on $D = 336°$, $I = -43°$ ($\alpha_{95} = 9°$) after cleaning and correction for tilt. Progressive demagnetization characteristically produced large initial direction change – Fig. 3(c) – and gradual intensity decrease – Fig. 2(b). Sample tilt correction of the principal group increased the precision of site means by almost five times, a fold test suggesting with 95 per cent confidence that the remanence is pre-Caradocian in age (McElhinny 1964: Cox 1969).

Steep positive and negative inclinations were found at the anomalous sites and af cleaning did not alter them appreciably – Fig. 3(a) and (b). Eight of the 11 anomalous sites were from the Haweswater dolerite or the volcanics intruded by it; two of these were unstable to af demagnetization. The other three were located close to sites having the directions of the principal group. The NRM intensities at the anomalous sites are comparable with those in the principal group. An attempt was made to combine the anomalous sites, assuming that both polarities occurred. However, the positive and negative groups are not closely antiparallel and the between-site precision was lower after cleaning and correction for tilt (Table 3). Therefore, the validity of this grouping must be very doubtful and it has to be conceded that the magnetization of these sites is unexplained.

The cones of 95 per cent confidence of the western and principal eastern groups overlap – Fig. 3(b) – and the test of Watson (1956a) shows that their mean stable remanence directions do not differ significantly. The combined mean direction is therefore taken as the result for the Borrowdale Volcanic Group. It is $D = 329°$, $I = -49°$ ($\alpha_{95} = 7°$), which corresponds to a palaeomagnetic pole at 0°, 203° E (Table 3 and Fig. 4). The significant improvement of precision on using dip-corrected site mean directions is convincing evidence that the stable remanence at all these sites predates the regional folding.

Discussion

The mean remanence direction of the Borrowdale Volcanic Group is significantly different at the 95 per cent confidence level (Watson 1956a) from that of the Eycott Group and the Carrock Fell Complex (Fig. 4). Comparing the Borrowdale Volcanics and the Eycott Group, the difference is one of declination only. However, because of the wide geographic extent of the sampling area and the appreciable stratigraphic thickness, the results from the Borrowdale Group probably represent a time span long enough to eliminate secular variation. The difference may therefore represent polar movement relative to the Lake District during Llanvirnian and Llandeilian time.

On the other hand, when the palaeomagnetic pole from the Borrowdale Volcanic Group is compared with other reliable Ordovician poles from the British Isles (Fig. 5), considerable overlap between many of the ovals of 95 per cent confidence is seen. Although not strictly applicable, a Watson–Irving (1957) two-tier analysis of the British Middle to Upper Ordovician poles plotted in Fig. 5 suggests that the between-Group dispersion is significant (F-ratio = 6.49 with degrees of freedom 324 and 16, hence between-group precision k_B = 29.8 and within-group precision $k_W = 8.9$). This would imply that the small differences between the various Group poles are significant. However, if the correct ages of magnetization have been assigned to each Group, it is impossible to construct a simple polar path for

Table 2. af cleaned results from the eastern Borrowdale Volcanics.

	Site	National Grid reference (100 km square NY)	Peak field (mT)	N	R	k	α_{95}	D (*in situ*)	I	D (dip corrected)	I
Principal Group	1 E	477085	30	6	4.94	58.2	16	345	−72	340	−23
	2 E	477084	30	6				not significant			
	3 E	459051	30	6	5.32	27.3	13	286	−68	316	−36
	4 E	425229	10	7	6.92	75.1	7	313	−66	330	−29
	5 E	425230	30	6				not significant			
	6 E	426230	50	5	4.76	16.7	19	11	−17	349	−43
	7 E	427230	30	6	5.86	36.8	11	9	−26	337	−48
	8 E	428231	50	5	4.91	43.4	12	19	−1	357	−23
	10 E	458051	40	6	5.97	145.1	6	275	−74	320	−46
	74 E	429217	20	7	6.38	9.6	21	348	−6	343	−33
	75 E	435216	20					not significant			
	76 E	386184	20	5	4.95	87.6	8	259	−67	312	−54
	77 E	386182	20	6	5.95	98.2	7	241	−82	332	−62
	78 E	386179	20	5	4.37	6.3	33	146	−81	3	−74
	79 E	386176	20	6	5.94	117.0	6	311	−81	340	−53
	80 E	387165	10	6	5.36	7.9	26	359	−60	355	−30
	81 E	405212	20	6	5.77	21.5	15	322	−37	322	−47
Anomalous results	9 E	458051	30	6	5.79	24.3	14	316	+53	248	+75
	11 E	459052	20	5	4.65	11.4	24	355	+70	146	+74
	67 E	495150				unstable to af demagnetization					
	68 E	496151				unstable to af demagnetization					
	69 E	497152	20	5	4.90	38.0	13	193	−82	125	−63
	70 E	497153	20	6	5.79	23.3	14	118	−56	114	−31
	71 E	498154	20	6	4.36	3.0	46	155	+73	248	+73
	72 E	499154	50	6				not significant			
	73 E	484152	20	6	5.58	12.0	20	249	−57	203	−71
	82 E	404214	20	6	5.70	16.7	17	248	−90	142	−65
	84 E	371210	30	5	4.71	13.6	22	284	+75	238	+58

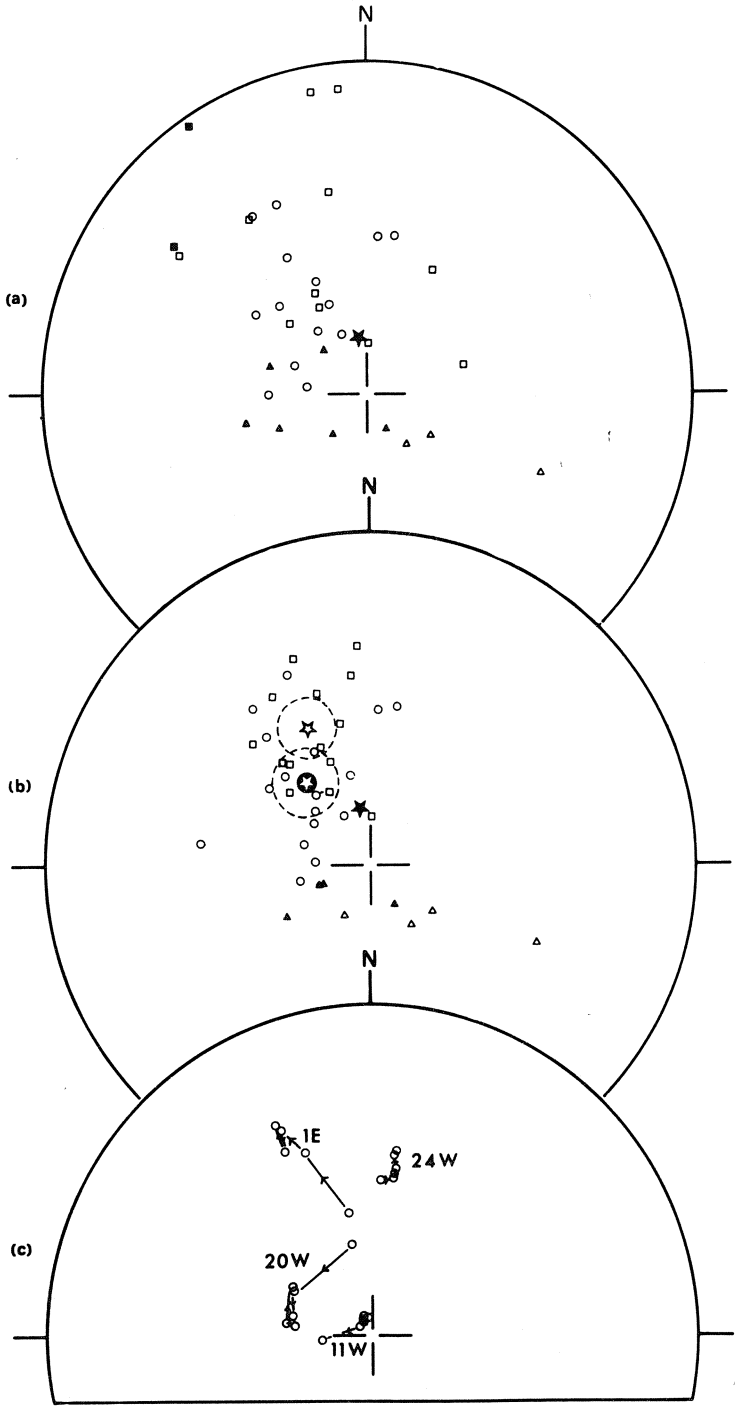


Figure 3. (a) and (b) dip-corrected directions of magnetization of sites in the Borrowdale Volcanic Group (a) before and (b) after af cleaning; (c) direction changes during progressive af demagnetization of specimens. Circles denote western sites, squares eastern sites forming the principal group, and triangles the anomalous eastern sites. The solid star is the direction of the present field. The open star is the mean direction for the principal group of eastern sites and the encircled star is the mean direction for the western sites. Throughout this paper open symbols denote upwards, solid symbols denote downwards, and broken circles are circles of 95 per cent confidence. Stereographic projections.

Table 3 Overall analysis of af cleaned results

							Palaeomagnetic pole			
	N	*R*	*k*	α_{95}	*D*	*I*	λ	φ	$d\psi$	$d\chi$
Western sites, *in situ*	17	14.33	6.0	16	45	−65				
Western sites, dip-corrected	17	15.67	12.2	11	322	−55	6S	208 E	12	16
Principal eastern sites. *in situ*	14	10.69	5.1	20	343	−57				
Principal eastern sites, dip-corrected	14	12.50	23.8	9	336	−43	7 N	198 E	8	11
All above sites, *in situ*	31	23.38	4.9	13	17	−64				
All above sites, dip-corrected	31	27.90	13.8	7	329	−49	0	203 E	7	9
Anomalous eastern sites, *in situ*	8	7.43	12.3	16	337	+80	71 N	334 E	30	32
Anomalous eastern sites, dip-corrected (assuming dual polarity)	8	7.31	10.2	18	284	+73	50 N	305 E	29	33

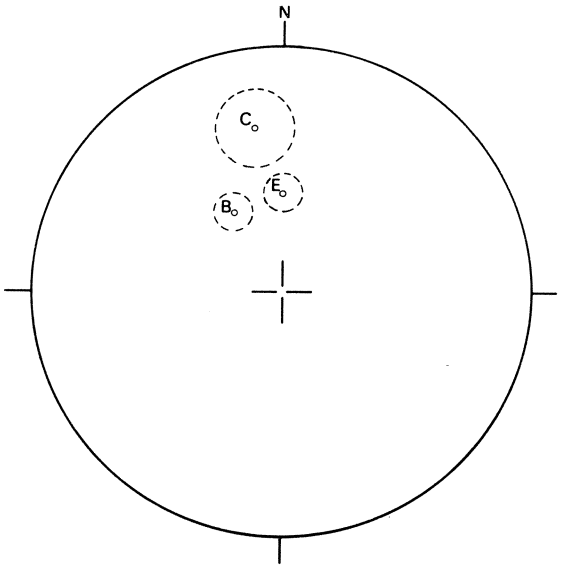


Figure 4. Comparison of the mean remanence directions of the Borrowdale Volcanic Group, B, the Eycott Group, E, and the Carrock Fell Complex, C. Stereographic projection.

the Middle to Upper Ordovician. Rather these apparent differences of pole position are to be attributed to failures in the sampling and statistical treatment of each study. Within the limits of error, then, the pole appears to have remained essentially stationary relative to the British Isles as a whole from Llanvirnian to Ashgillian time.

In the absence of polar wander or a distinctive polarity reversal sequence during this period, it would seem that palaeomagnetic results cannot be used to assist in correlating volcanic strata across the Lake District. Even if progressive polar movement did occur there are many serious limitations to the application of this method. Several are connected with

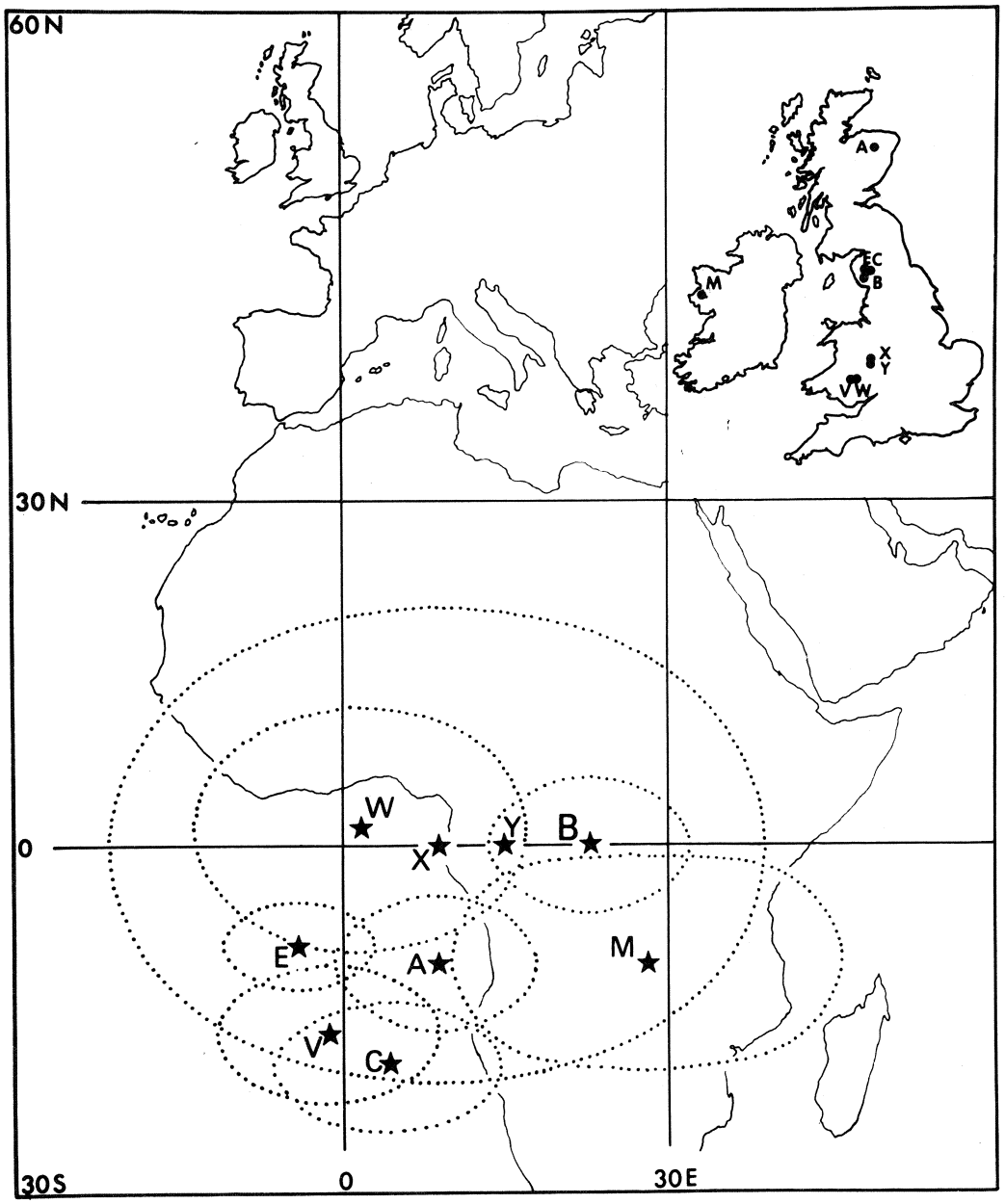


Figure 5. Ordovician palaeomagnetic poles for the British Isles with approximate ovals of 95 per cent confidence. A – Aberdeenshire Gabbros, B – Borrowdale Volcanic group, C – Carrock Fell Complex, E – Eycott Group, M – Mweelrea Ignimbrites, V – Builth Volcanics, W – intrusions of the Builth Inlier, X – Breidden Hill extrusives, Y – Breidden Hill intrusives. Mercator projection.

sampling. There is limited outcrop of suitable uncleaved flows, and sampling lavas erupted in a short time interval may not eliminate geomagnetic secular variation. Accurate tilt correction is important but cannot always be made. As stated earlier, the local dip for the andesites sampled was measured on nearby bedded tuffs; it is assumed in all cases that the lavas were tilted by the same amount as the tuffs although the tuffs are usually cleaved and therefore

suffered greater tectonic strain. Tilt correction was carried out by simple rotation about the strike, making no allowance for possible plunge (although in the Borrowdale Volcanics this is usually less than 10° and neglecting it leads to small errors). It has to be assumed that the lavas were erupted on to a horizontal surface, since depositional dip can rarely be determined in the presence of tectonic dip. Statistical difficulties also arise. There are uncertainties associated with the values of α_{95} for groups of sites whose site means do not form a strictly Fisherian distribution, and in order to apply statistical tests for significance of differences between mean directions, each should have the same precision. Finally it is not clear whether samples, sites, or even lava flows constitute independent observations and hence what is the proper level at which to test for significance. All these factors make it impossible to resolve small palaeomagnetic differences between groups of stratigraphically equivalent sites.

An overall Ordovician pole position has been calculated by combining the result from the Borrowdale Volcanic Group with the other British Ordovician poles plotted in Fig. 5. They are from the Aberdeenshire Gabbros, the Eycott Group, the Builth Volcanics, the Mweelrea Ignimbrites, and intrusions of Ashgillian age in the Builth Inlier (Briden *et al.* 1973), from intrusive and extrusive rocks of Breidden Hill (recalculated from Piper & Stearn 1975), and a modified result from the Carrock Fell Complex. Equal weights were given to sites in Table 1 of Piper & Stearn for the intrusive and extrusive rocks respectively. The Carrock Fell result has been updated from that presented by Briden & Morris (1973), and is now derived from 17 sites compared with 11 formerly, as a result of additional work by Morris (1973). Moreover, a different tilt correction has been applied based on the assumption that the overlying Drygill Shales were in a symmetrical syncline at the time of emplacement of the Carrock Fell Complex. Using this assumption brings the mineral lamination of the Complex close to the vertical and the pole position nearer to the Ordovician mean than was the original published pole from this Complex. The best estimate of the stable remanence direction for the Carrock Fell Complex is now $D = 351°$, $I = -29°$ ($\alpha_{95} = 10°$). The corresponding pole position is 19° N, 184° E, and the inference is that minor tilting of the Complex took place after its intrusion in essentially a vertical form. Giving equal weight to each Ordovician palaeomagnetic pole from the British Isles, it is concluded that the overall Ordovician pole was in the vicinity of 7° N, 190° E ($\alpha_{95} = 9°$).

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